How to read a paper (Part II)

Tao Du

Recap: The logic behind a scientific paper

- What is the research problem?
  - Is it an important problem, and is it of interest to the scientific community?
- What are the state of the arts (SOTAs)?
  - What are their limitations? Why is it necessary to develop a new method?
- What is the main method?
  - Describe the technical details behind it.
- What are the results?
  - Show evidence of the method outperforming SOTAs on certain metrics.

Recap: The logic behind a scientific paper

- What is the research problem? -> **Introduction**
  - Is it an important problem, and is it of interest to the scientific community?
- What are the state of the arts (SOTAs)? -> **Related Work**
  - What are their limitations? Why is it necessary to develop a new method?
- What is the main method? -> **Method**
  - Describe the technical details behind it.
- What are the results? -> **Experiments & Discussions**
  - Show evidence of the method outperforming SOTAs on certain metrics.

Recap: A recommended reading order

Abstract

-> **Intro** -> **Related Work**

-> **Experiments & Discussions**

-> Methods

-> Conclusions

How to read an experiment

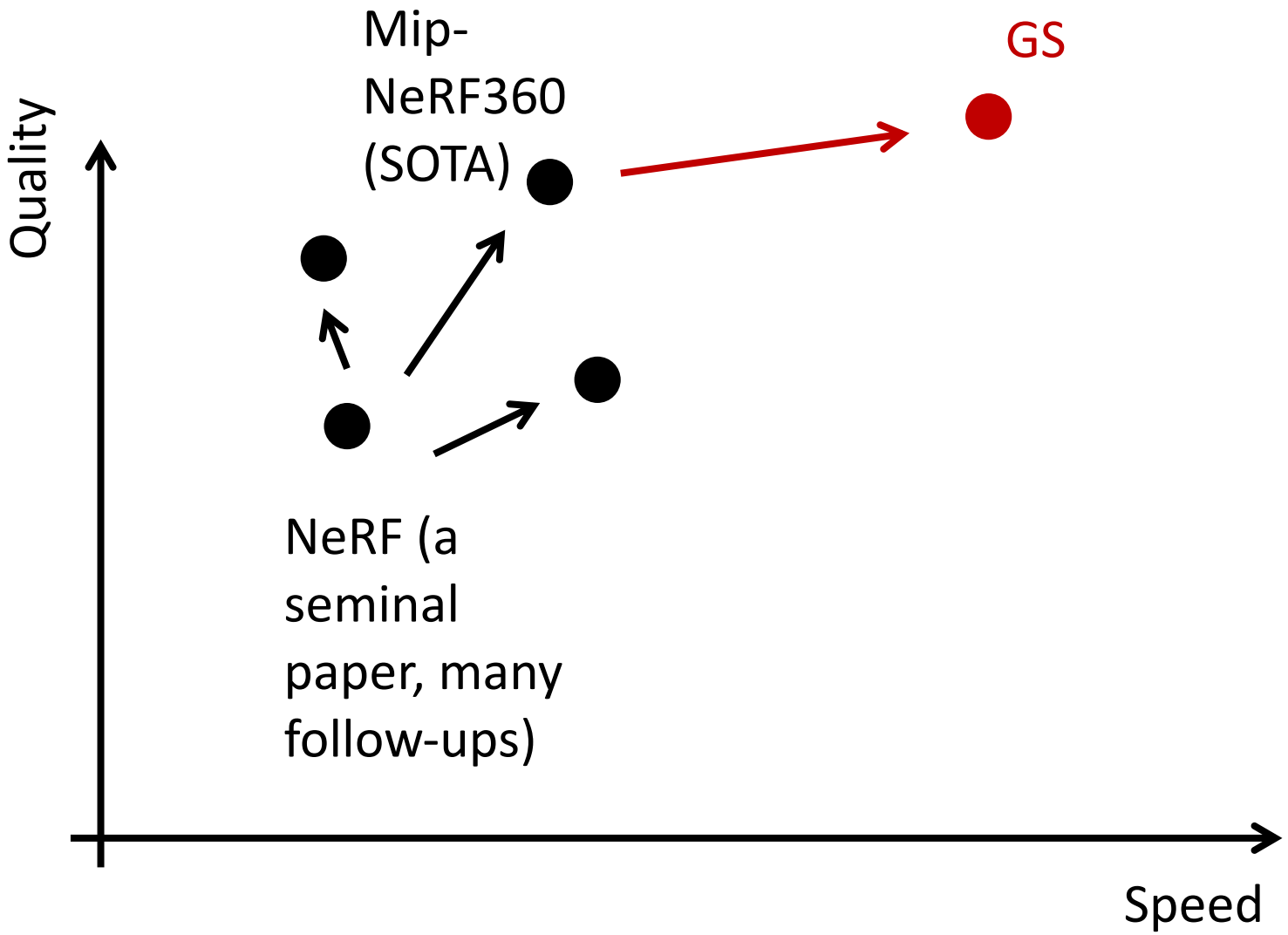
The logic behind an experiment

Goal: “This method is better than Method M (SOTA) on Benchmark B.”

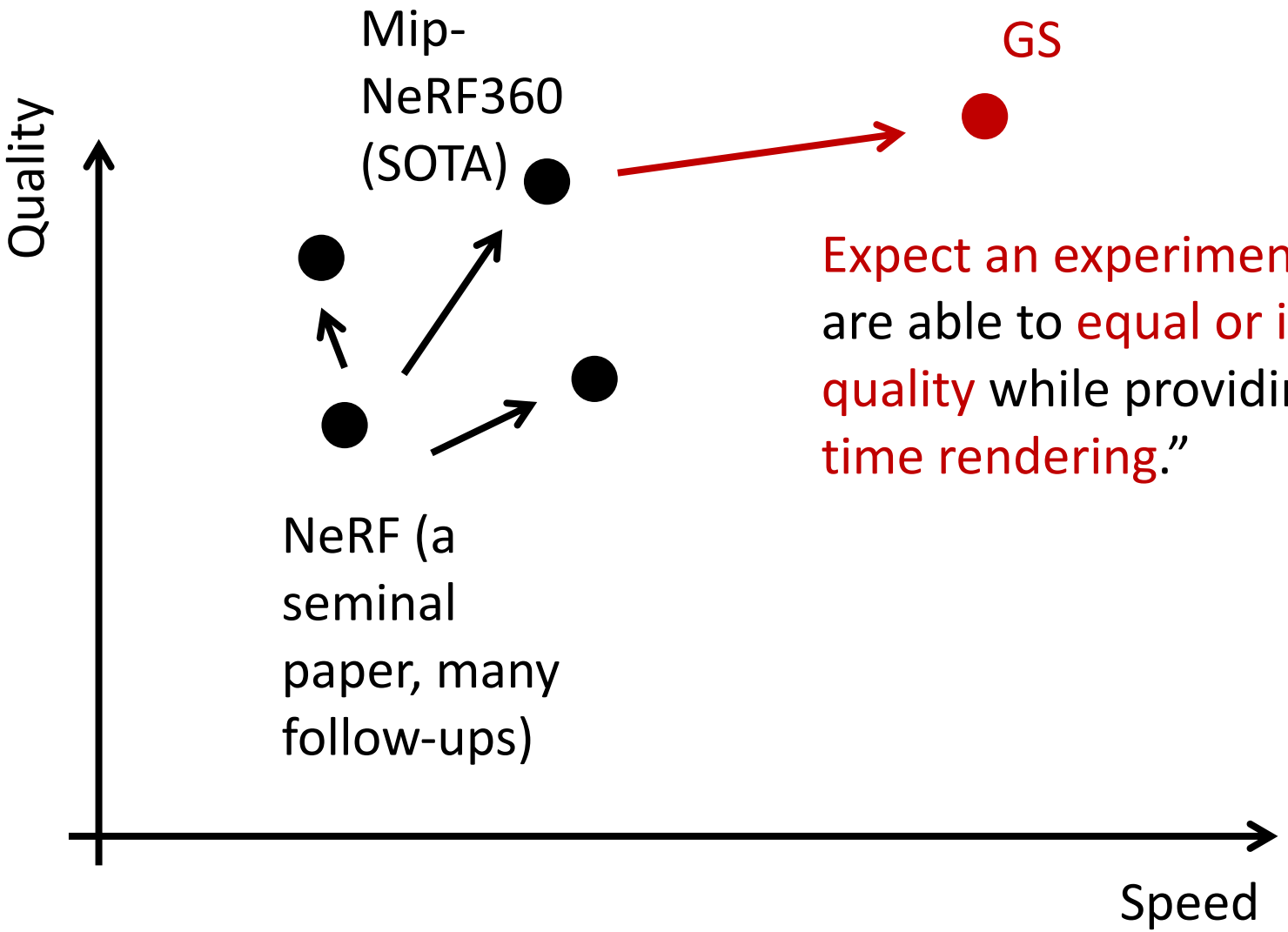
So an experiment needs to provide the following context:

- The benchmark B;
- The SOTA method M;
- Some performance metrics.

Example: Gaussian Splatting



Example: Gaussian Splatting



Expect an experiment to prove this arrow: “we are able to equal or in some cases surpass this quality while providing fast training and real-time rendering.”

Example: Gaussian Splatting (Sec. 7.2)

“We tested our algorithm on a total of 13 real scenes taken from previously published datasets and the synthetic Blender dataset [Mildenhall et al. 2020]. In particular, we tested our approach on the full set of scenes presented in Mip-Nerf360 [Barron et al. 2022], which is the current state of the art in NeRF rendering quality, two scenes from the Tanks&Temples dataset [2017] and two scenes provided by Hedman et al. [Hedman et al. 2018]...”

Example: Gaussian Splatting (Sec. 7.2)

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Example: Gaussian Splatting

Benchmark	Novel-view synthesis on 13 real scenes: Mip-Nerf360, Tanks & Temples, and Hedman et al.
SOTA	
Metrics	

Example: Gaussian Splatting (Sec. 7.2)

"…In terms of quality, the current state-of-the-art is Mip-Nerf360 [Barron et al. 2021]. We compare against this method as a quality benchmark. We also compare against two of the most recent fast NeRF methods: InstantNGP [Müller et al. 2022] and Plenoxels [Fridovich-Keil and Yu et al. 2022].

We use a train/test split for datasets, using the methodology suggested by Mip-NeRF360, taking every 8th photo for test, for consistent and meaningful comparisons to generate the error metrics, using the standard PSNR, L-PIPS, and SSIM metrics used most frequently in the literature; please see Table 1…"

Example: Gaussian Splatting (Sec. 7.2)

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Example: Gaussian Splatting

Benchmark	Novel-view synthesis on 13 real scenes: Mip-Nerf360, Tanks & Temples, and Hedman et al.
SOTA	Mip-Nerf360, InstantNGP, Plenoxels, …
Metrics	PSNR, L-PIPS, and SSIM, …

“please see Table 1…” -> **Discuss the possible layout of Table 1.**

Example: Gaussian Splatting (quantitative)

The actual Table 1 in GS:

Dataset	Mip-NeRF360					
Method|Metric	$SSIM^{\uparrow}$	$PSNR^{\uparrow}$	$LPIPS^{\downarrow}$	Train	FPS	Mem
Plenoxels	0.626	23.08	0.463	25m49s	6.79	2.1GB
INGP-Base	0.671	25.30	0.371	5m37s	11.7	13MB
INGP-Big	0.699	25.59	0.331	7m30s	9.43	48MB
M-NeRF360	$0.792^{\dagger}$	$27.69^{\dagger}$	$0.237^{\dagger}$	48h	0.06	8.6MB
Ours-7K	0.770	25.60	0.279	6m25s	160	523MB
Ours-30K	0.815	27.21	0.214	41m33s	134	734MB

Example: Gaussian Splatting (quantitative)

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Example: Gaussian Splatting (quantitative)

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SOTA methods

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Example: Gaussian Splatting (quantitative)

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Metrics (quality)

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Example: Gaussian Splatting (quantitative)

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HIGHER is better

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Example: Gaussian Splatting (quantitative)

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LOWER is better

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Top-3 performances are colored

Example: Gaussian Splatting (quantitative)

The actual Table 1 in GS:

Metrics (speed)

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Orders-of-magnitude faster

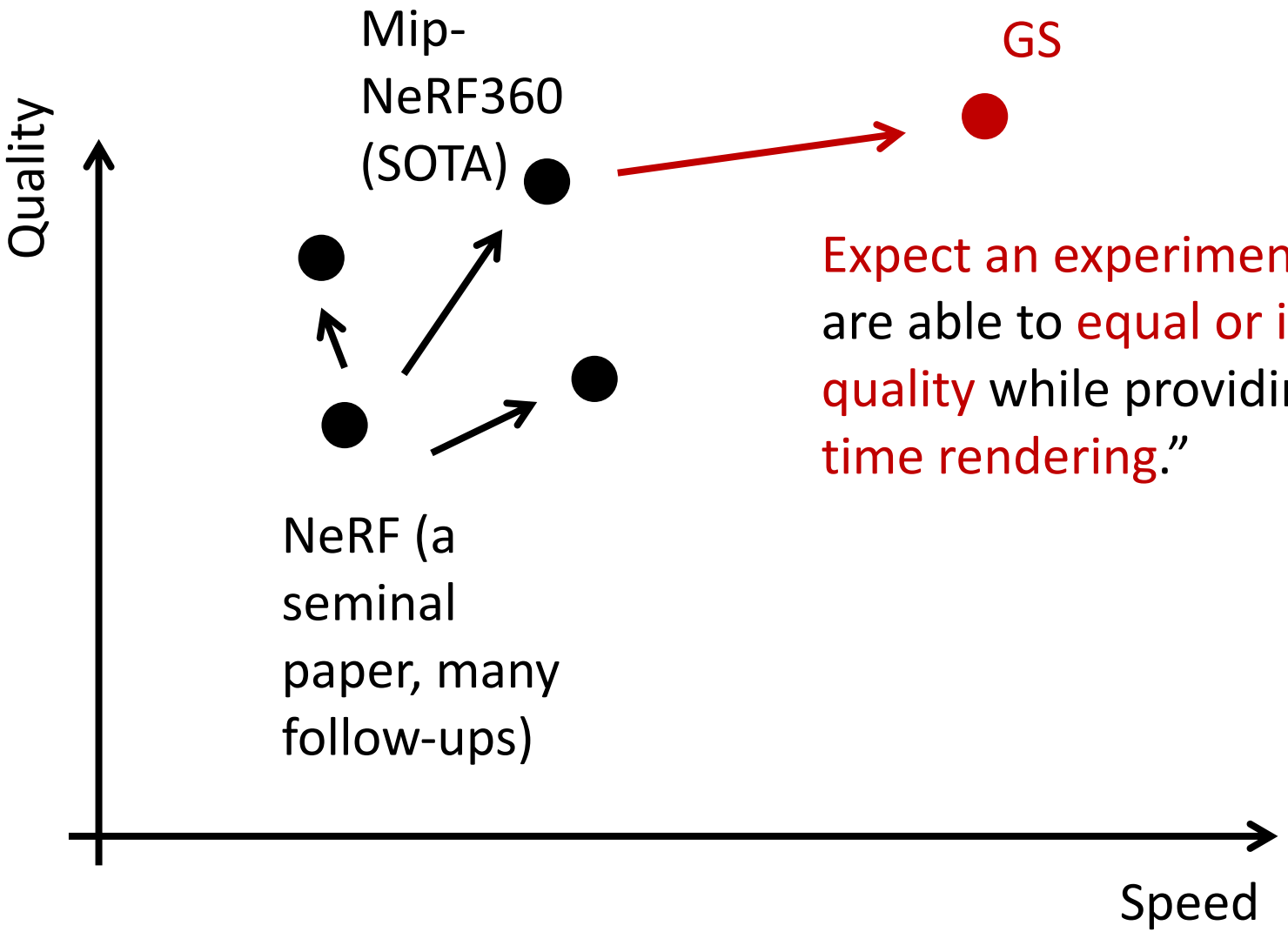
Example: Gaussian Splatting (Sec. 7.2)

“The table shows that our fully converged model achieves quality that is on par and sometimes slightly better than the SOTA Mip-NeRF360 method; note that on the same hardware, their average training time was 48 hours2, compared to our 35-45min, and their rendering time is 10s/frame. We achieve comparable quality to InstantNGP and Plenoxels after 5-10m of training, but additional training time allows us to achieve SOTA quality which is not the case for the other fast methods…”

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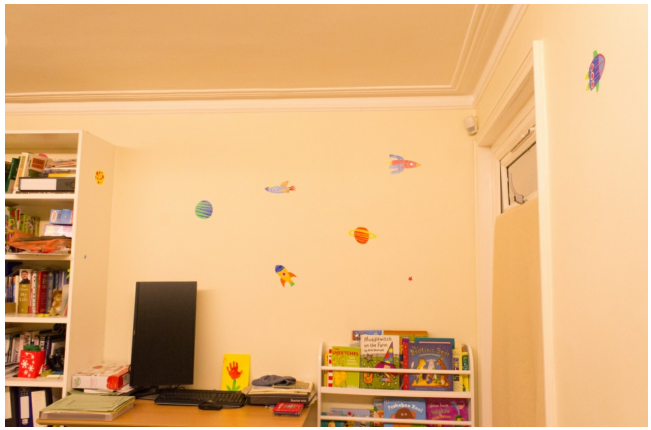
Expect an experiment to prove this arrow: “we are able to equal or in some cases surpass this quality while providing fast training and real-time rendering.”

Example: Gaussian Splatting (qualitative)

The problem setup of novel view synthesis:

Input: photos taken by cameras from different viewpoints.

Output: a model that returns realistic images from novel views (new camera locations and poses)

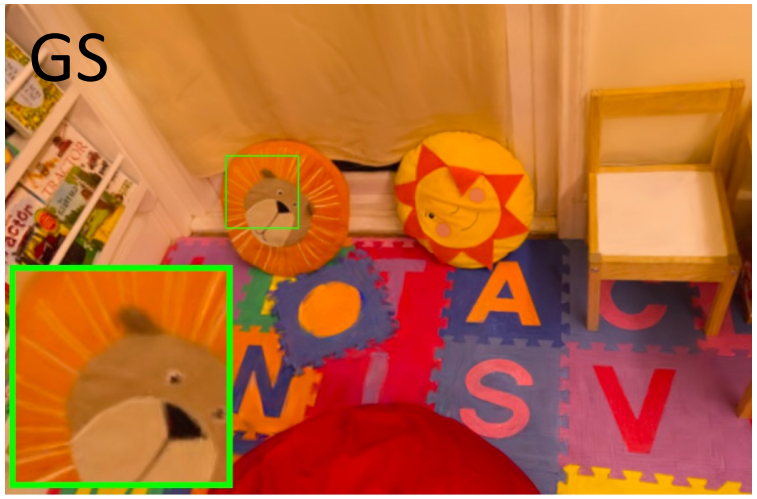


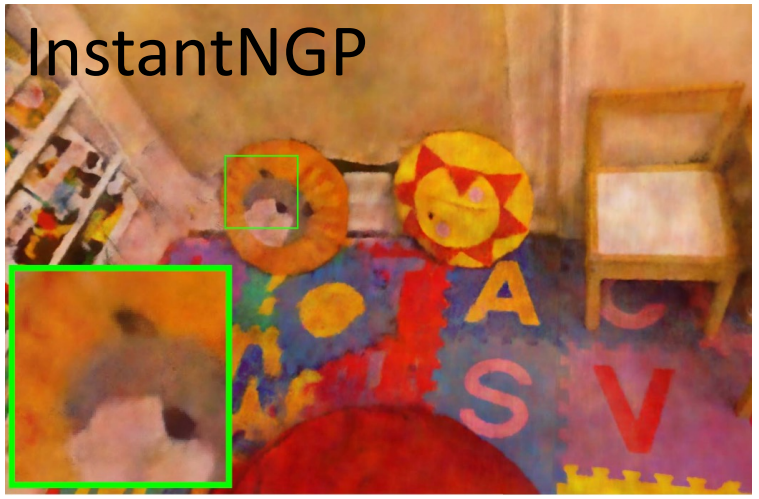
…

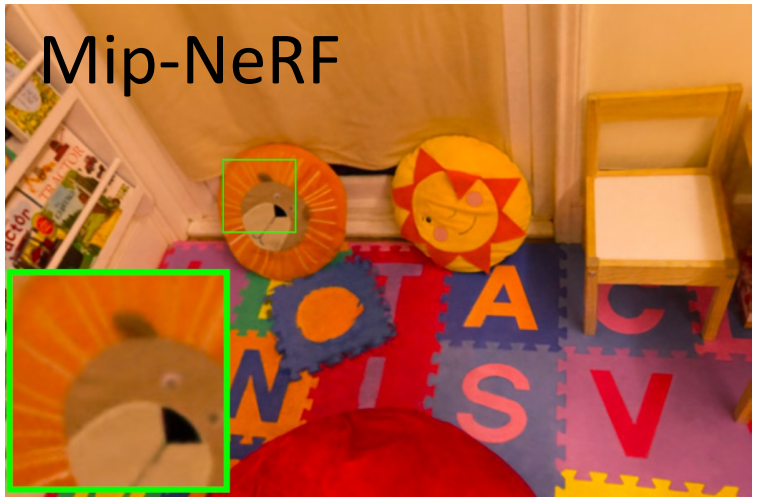

Input: 225 images in this example

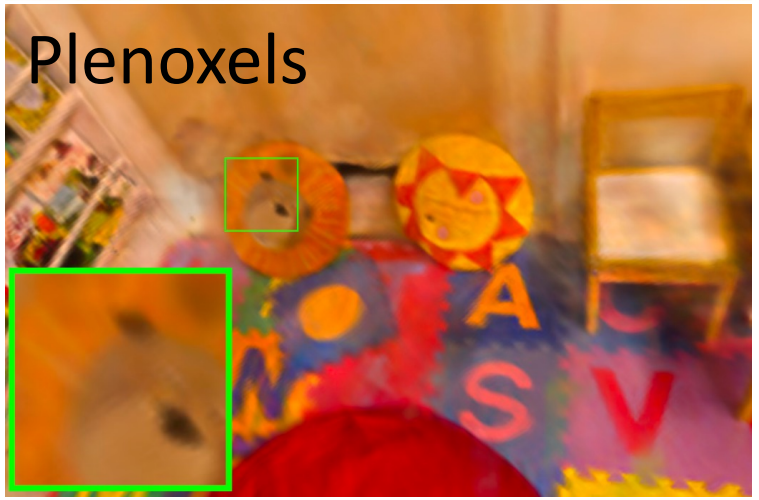

Menu
Views
Capture
▶ 3D Gaussians
▶ Camera Point view
Musical Instruments
▼ Metrics
89.37 (11.19 ms)

Example: Gaussian Splatting (qualitative)









Example: Gaussian Splatting (Sec. 7.2)

“We also show visual results of this comparison for a left-out test view for ours and the previous rendering methods selected for comparison in Fig. 5; the results of our method are for 30K iterations of training. We see that in some cases even Mip-NeRF360 has remaining artifacts that our method avoids (e.g., blurriness in vegetation – in Bicycle, Stump – or on the walls in Room). In the supplemental video and web page we provide comparisons of paths from a distance. Our method tends to preserve visual detail of well-covered regions even from far away, which is not always the case for previous methods.”

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More video results on the next few slides.


Menu Views Capture
3D Gaussians
Camera Point view
Metrics
57.56 (17.37 ms)
VSync On


Menu Views Capture
▶ 3D Gaussians
▶ Camera Point view
▼ Metrics
59.97 (16.67 ms)
VSync On

Summary: how to read a paper

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  - What are their limitations? Why is it necessary to develop a new method?
- What is the main method? -> **Method**
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Summary: how to read a paper

In general, a good paper tells a coherent story:

- Introduction identifies a research gap;
- Related work confirms the gap hasn't been filled by previous work
- Method explains how this gap is filled (requires domain-specific knowledge; not covered in this tutorial);
- Results prove this gap has been filled.

(There are exceptions, though, e.g., some theory papers)